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Roll No

EE-503 (GS)

B.E. V Semester Examination, June 2020

Grading System (GS)

Signals and Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume any missing data.

1. a) Explain the following with respect to systems:
 i) Causality ii) Linearity
 iii) Stability
 b) What are linear Time Invariant systems? How an LTI system is realized?

2. What do you mean by the convolution? Explain it with suitable example and it's properties.

3. Determine if the system described below are LTI or not.
 i) $y(n) = nx(n)$ ii) $y(n) = x(n^2)$
 iii) $y(n) = x^2(n)$ iv) $y(n) = e^{x(n)}$

4. What is the basic difference between laplace and Fourier Transform? What is the purpose of laplace Transform?

5. Write short notes on any two of the following:
 a) DFT b) Properties of DFT
 c) Significance of poles and zero's

6. Determine the Nyquist rate of the following signals:
 i) $x(t) = 1 + \cos(2000\,\pi t) + \sin(4000\,\pi t)$
 ii) $x(t) = \dfrac{\sin(4000\pi t)}{\pi t}$

7. a) what is sampling theorem? Define aliasing effect.
 b) Write a note on 'Sampling of discrete-time signals'.

 OR

 What is energy signal? State and prove the Parseval's theorem for energy signals.

8. Write short notes on (any two):
 i) Singularity functions. ii) Region of convergence.
 iii) Frequency response of LTI systems. iv) Sampling in frequency domain.

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